Asymptotic Analysis of Machine Learning Training Algorithms

How Big O Applies to Model Training

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Abstract

As machine learning models continue to grow in size and complexity, understanding the computational cost of training has become increasingly important. Although optimization algorithms such as Gradient Descent, Stochastic Gradient Descent (SGD), and Adam are widely used, their training efficiency is often discussed separately from the principles of asymptotic analysis taught in Design and Analysis of Algorithms (DAA). This paper examines how Big O notation can be used to analyze and compare the scalability of machine learning training algorithms through a unified computational perspective. The study investigates how factors such as dataset size, model parameters, batch size, and training epochs influence the overall training cost of different optimization methods. Rather than introducing a new optimization algorithm, this research presents a unified educational complexity estimation formulation that combines these variables into a single analytical model for predicting training complexity before model execution. The proposed approach aims to bridge the gap between theoretical

algorithm analysis and real-world machine learning training, making complexity estimation more intuitive for students, researchers, and practitioners. By connecting classical asymptotic analysis with modern optimization techniques, this work provides a structured foundation for understanding how training algorithms scale as machine learning systems become increasingly larger.

Keywords - Asymptotic Analysis, Big O Notation, Machine Learning, Training Algorithms, Gradient Descent, Stochastic Gradient Descent, Time Complexity, Design and Analysis of Algorithms

I. Introduction

Machine learning has become a fundamental technology in modern computing, supporting applications such as medical diagnosis, recommendation systems, fraud detection, autonomous vehicles, and natural language processing. As these applications become more data-intensive, training machine learning models requires increasingly larger datasets and more complex model architectures. This growth has made computational efficiency an essential consideration when selecting training methods, as training time and resource requirements can increase significantly with problem size.

In Design and Analysis of Algorithms (DAA), asymptotic analysis provides a mathematical framework for studying how an algorithm's computational cost changes as its input grows. Big O notation allows algorithms to be compared based on their growth rate rather than hardware-dependent execution time, making it one of the most widely used tools for evaluating algorithm scalability. Although these principles are routinely applied to classical algorithms such as sorting and searching, their practical application to machine learning training is often presented in a fragmented manner across different optimization techniques.

Machine learning training relies on optimization algorithms such as Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch SGD, Momentum, RMSProp, and Adam, each of which follows a different strategy for updating model parameters. While these algorithms have been extensively studied for convergence speed and optimization performance, fewer studies explain their computational behaviour through a unified asymptotic perspective that is accessible to both algorithm students and machine learning practitioners.

This research addresses that need by examining how key training variables—including dataset size, model parameters, batch size, and training epochs—influence computational complexity. Instead of proposing a new optimization algorithm, the study presents a Unified Big O-Based Complexity Estimation Formulation that provides a structured way to estimate training cost before model execution. By connecting classical asymptotic analysis with practical machine learning training, this work offers a clearer understanding of how optimization algorithms scale as modern machine learning systems continue to grow.

A. Research Objectives

The primary objective of this research is to analyse the asymptotic time complexity of major machine learning training algorithms and develop a practical framework for estimating training complexity.

The specific objectives are:

- To study optimization algorithms commonly used for machine learning training.
- To derive the asymptotic time complexity of Gradient Descent, SGD, Mini-batch SGD, Momentum, RMSProp, and Adam.
- To compare the computational efficiency of these optimization algorithms.
- To identify the factors that influence training complexity.
- To propose a unified Big O-based complexity estimation framework.
- To validate theoretical complexity through experimental analysis.

II. Literature Review

The computational cost of training machine learning models has become increasingly important as datasets and model sizes continue to grow. Earlier research mainly focused on improving prediction accuracy, but recent studies have shifted attention toward training efficiency, scalability, and optimization behaviour. This evolution shows that understanding how training algorithms grow with increasing data and model size is now as important as improving their accuracy. From a Design and Analysis of Algorithms (DAA) perspective, asymptotic analysis provides a systematic way to study this growth and compare different optimization methods.

A. Early Foundations of Optimization

The foundation of modern machine learning training lies in gradient-based optimization, where model parameters are repeatedly updated to minimize a loss function. Early optimization methods relied on processing the entire training dataset during every update, which worked well for small datasets but became computationally expensive as data volumes increased. The introduction of stochastic optimization changed this approach by allowing parameter updates using individual samples or small subsets of data, making training more practical for large-scale problems.

This transition established an important idea that remains relevant today: reducing computation per update can significantly improve training efficiency without changing the overall learning objective.

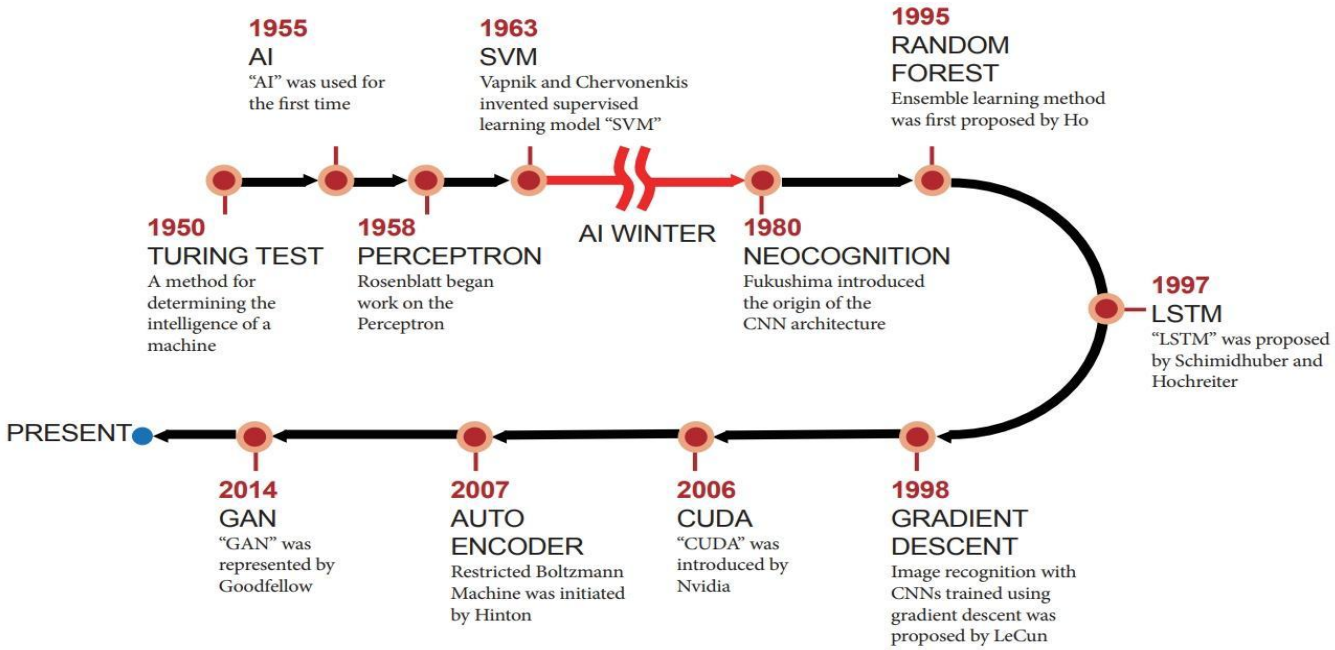


Figure 1. Evolution of optimization algorithms used in machine learning training.

Figure 1 illustrates the evolution of optimization algorithms from classical gradient descent to modern adaptive optimizers.

B. Optimization for Large-Scale Machine Learning

As machine learning applications expanded, researchers recognized that traditional optimization techniques could not efficiently handle millions of training samples. Large-scale optimization studies demonstrated that Stochastic Gradient Descent (SGD) became the preferred choice because it reduced computational overhead while maintaining effective convergence.

Another important development during this period was the adoption of Mini-Batch SGD, which processes small groups of training samples instead of either a single sample or the complete dataset. This approach balances computational efficiency with training stability and has become the standard optimization strategy for deep learning.

Although these studies clearly explained why stochastic methods outperform full-batch optimization on large datasets, they primarily focused on optimization performance rather than providing a unified mathematical framework for estimating training complexity before execution.

C. Evolution of Modern Training Algorithms

As neural networks became deeper and more complex, optimization research moved beyond basic gradient descent. New algorithms such as Momentum, RMSProp, and Adam were introduced to improve convergence speed and training stability by incorporating historical gradient information or adaptive learning rates.

These optimizers demonstrated that training performance depends not only on the number of computations performed but also on how efficiently parameter updates are executed. While many of these methods retain similar asymptotic time complexity per update, they differ in memory usage and convergence behavior, making optimizer selection an important practical consideration.

D. Theoretical Understanding of Training Complexity

More recent research has explored optimization from a mathematical perspective rather than focusing only on implementation. Studies on deep learning optimization have examined why large neural networks remain trainable despite highly complex loss landscapes, while other work has modelled gradient-based optimization using continuous mathematical systems to better understand long-term convergence behaviour.

These theoretical developments strengthened the mathematical understanding of optimization by connecting algorithm behaviour with convergence theory. However, many of these approaches rely on advanced mathematical models that are difficult to use for practical complexity estimation, especially for students and practitioners who need an intuitive understanding of computational growth.

E. Current Trends in Machine Learning Training

Current research shows that training complexity is influenced by several interacting factors rather than dataset size alone. Model parameter count, batch size, hardware architecture, memory usage, and distributed computation all affect the overall training process.

This broader perspective reflects the changing nature of machine learning systems, where computational efficiency depends on both algorithmic design and implementation strategy. Nevertheless, Big O notation remains valuable because it provides a hardware-independent way to compare how algorithms scale as these factors increase

F. Comparative Analysis of Existing Studies

The reviewed studies collectively demonstrate the evolution of optimization techniques from classical gradient-based methods to modern adaptive optimizers. While each approach offers distinct advantages, they also exhibit limitations that motivate the need for a more unified complexity estimation framework.

Research Area	Major Contribution	Limitation
Full-Batch Gradient Descent	Stable optimization	High computational cost for large datasets
Stochastic Gradient Descent	Faster updates with reduced computation	Higher gradient noise
Mini-Batch SGD	Balance between speed and stability	Requires batch-size tuning
Adaptive Optimizers (Adam, RMSProp)	Faster convergence with adaptive updates	Additional memory overhead
Deep Learning Optimization Theory	Strong mathematical understanding	Limited practical complexity estimation

The comparison highlights that existing research has successfully improved optimization efficiency but has generally treated convergence behaviour and computational complexity as separate areas of study.

Research Gap

The reviewed studies have significantly improved optimization algorithms and their theoretical understanding, but they generally examine convergence, optimizer behaviour, or mathematical analysis separately. Few studies provide a practical framework that combines dataset size, batch size, model parameters, and training epochs into a single asymptotic model for estimating training complexity before execution.

This research addresses that gap by developing a unified Big O-based complexity estimation framework that connects the principles of Design and Analysis of Algorithms (DAA) with the practical behaviour of modern machine learning training algorithms. Rather than proposing another optimizer, the study focuses on making complexity analysis more structured, interpretable, and applicable to real-world training scenarios.

III. Research Methodology

This study follows a comparative analytical research approach to examine how asymptotic analysis can be applied to machine learning training algorithms. The objective is not to develop a new optimization algorithm, but to analyse the computational behaviour of existing training methods and build a practical framework for estimating their training complexity using Big O notation.

The research was carried out in four stages: reviewing existing optimization methods, identifying the factors that affect training complexity, deriving the asymptotic complexity of selected algorithms, and developing a unified complexity estimation model.

A. Research Design

The research focuses on six widely used training algorithms because they represent the progression from classical optimization to modern adaptive optimization.

Algorithm	Purpose
Gradient Descent	Baseline optimization
Stochastic Gradient Descent	Sample-based optimization
Mini-Batch SGD	Practical deep learning training
Momentum	Faster convergence
RMSProp	Adaptive learning
Adam	Modern optimization standard

These algorithms were selected because they are commonly used in supervised learning and deep learning while exhibiting different computational characteristics.

B. Complexity Analysis Framework

To analyse each algorithm consistently, the training process was divided into four fundamental operations.

1. Forward propagation
2. Loss calculation
3. Backpropagation
4. Parameter update

The computational cost of these operations was expressed using Big O notation, and the overall training complexity was derived by considering repeated execution across batches and training epochs.

C. Parameters Used in the Analysis

Instead of representing complexity with a single input size, this study considers multiple variables that directly influence training cost.

Symbol	Description
n	Number of training samples
d	Number of input features
p	Number of trainable parameters
b	Batch size
e	Number of training epochs

These variables remain consistent throughout the mathematical analysis and algorithm comparison.

D. Proposed Analysis Approach

After comparing the selected algorithms, this research develops a unified complexity estimation model that relates dataset size, batch size, training epochs, and model parameters within a single analytical expression.

$$T = O\left(e \times \frac{n}{b} \times p\right)$$

This framework serves as the foundation for the complexity analysis presented in the next section and provides a practical method for estimating training cost before model execution.

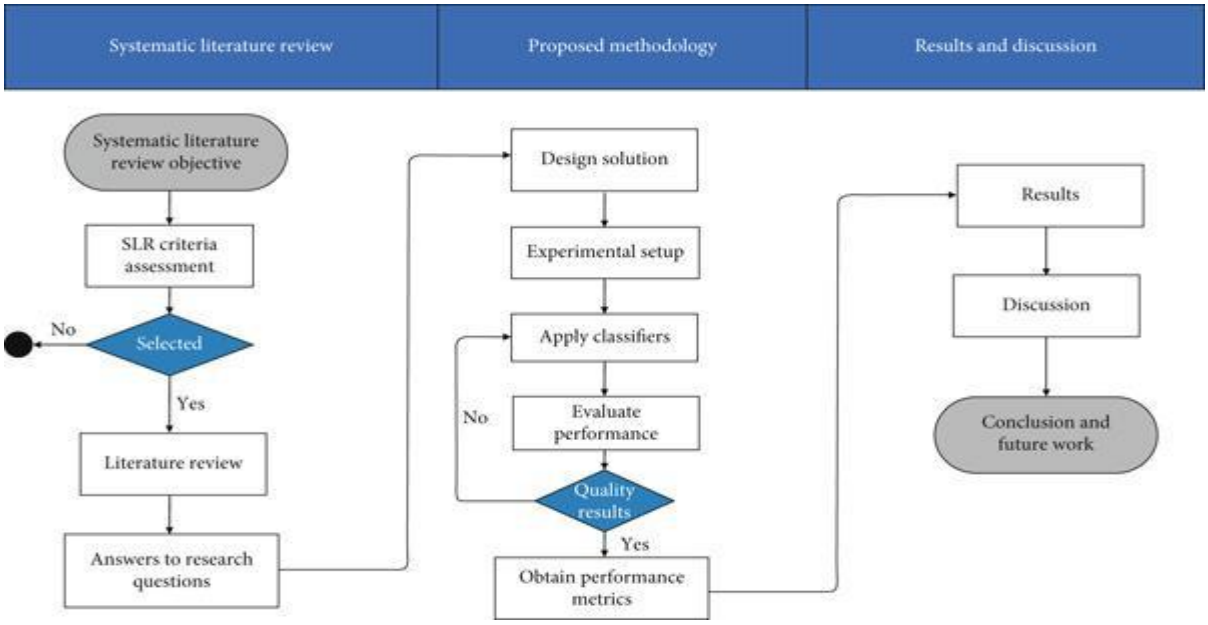


Figure 2. Research methodology adopted in this study.

Figure 2 summarizes the research methodology adopted in this study.

IV. Mathematical Foundation

Asymptotic analysis provides a mathematical way to study how the computational cost of an algorithm changes as the input size increases. In machine learning, training is not a single operation but a repeated sequence of computations performed across multiple iterations. Every training cycle involves processing input data, calculating prediction errors, computing gradients, and updating model parameters. As the dataset and model size grow, the number of these operations also increases, making asymptotic analysis an appropriate tool for evaluating training scalability.

Unlike many classical algorithms that depend on a single input variable, machine learning training is influenced by several interconnected factors. The overall computational cost depends on the amount of training data, the number of model parameters, the chosen batch size, and the number of training epochs. Understanding the relationship between these variables forms the mathematical basis for the complexity analysis presented in this study.

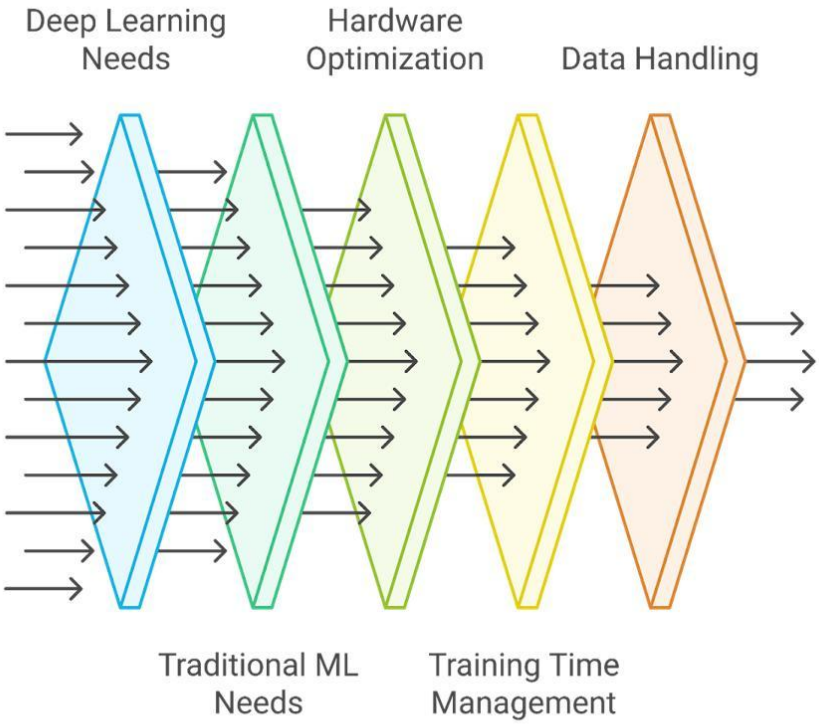


Figure 3 . Multi-Stage Framework for Machine Learning Training Optimization

This figure visually introduces the variables before they are defined mathematically, making the section easier to understand.

A. Variables Used in the Analysis

To maintain consistency throughout the paper, the following symbols are used in all mathematical derivations.

Symbol	Description
n	Number of training samples
d	Number of input features
p	Number of trainable parameters
b	Batch size
e	Number of training epochs

These variables represent the primary factors that influence the computational growth of machine learning training algorithms.

B. Role of Big O Notation in Training Analysis

Big O notation describes the upper limit of an algorithm's growth as the problem size increases. Rather than measuring execution time on a particular computer, it focuses on how the number of required operations grows when the input becomes larger.

For example, if an algorithm performs approximately $3n + 5$operations, its growth is represented as $O(n)$because the linear component dominates for large values of n. The same principle applies to machine learning training, where repeated computations become more significant than constant initialization costs.

In this research, Big O notation serves as the foundation for comparing different optimization algorithms and estimating their training complexity using a common analytical perspective.

Figure 4 illustrates the hierarchical relationship between training variables and computational complexity.

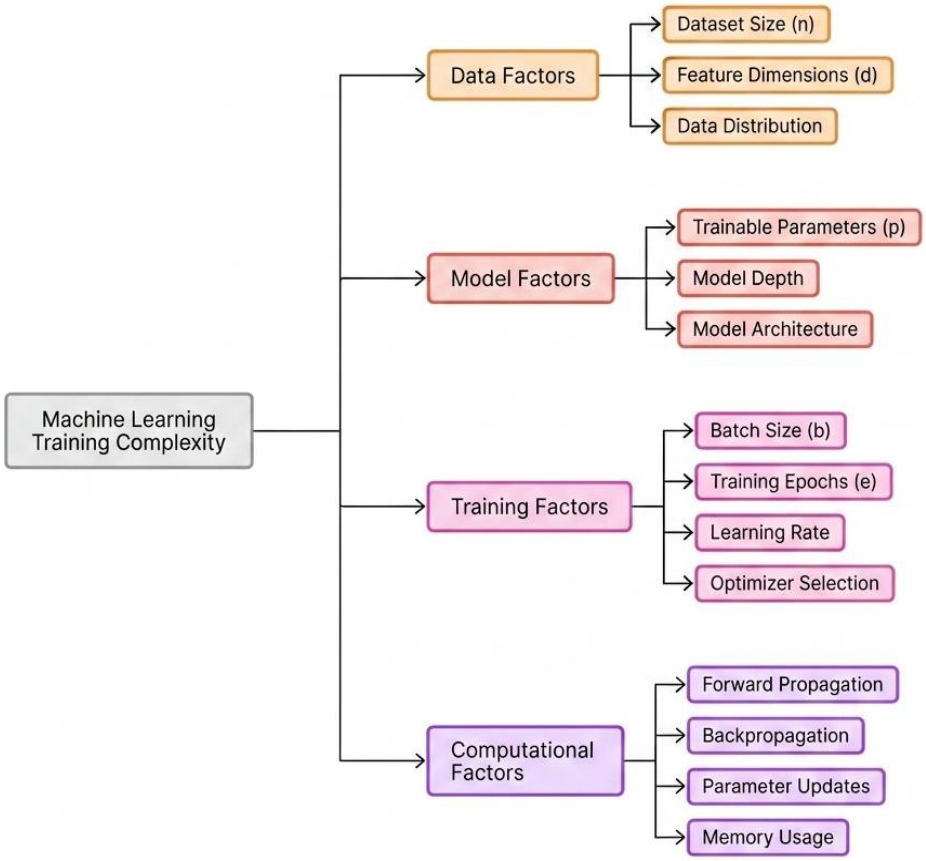


Figure 4. Hierarchical Framework for Machine Learning Training Complexity

V. Complexity Analysis of Machine Learning Training Algorithms

The training process of a machine learning model consists of repeated computations that continue until the model reaches an acceptable level of accuracy. Although different optimization algorithms follow different update strategies, they all perform the same fundamental cycle: processing training data, calculating prediction error, computing gradients, and updating model parameters. The difference lies in how frequently these operations are performed and how much data is processed during each update.

To understand how training cost grows, each algorithm is analyzed using the same computational perspective, allowing a fair comparison of their scalability.

A. Gradient Descent

Per iteration complexity: O(nd)

Overall training complexity after e epochs: O(end)

Gradient Descent updates model parameters only after processing the entire training dataset. During one iteration, every training sample contributes to the gradient calculation before a single update is performed.

If a dataset contains n samples and each sample has d features, the algorithm performs approximately n × d computational operations during one complete iteration.

Therefore, the time complexity for one iteration is

$$O(nd)$$

After training for e epochs, the overall complexity becomes

$$O(end)$$

Gradient Descent provides stable updates but becomes computationally expensive for large datasets because every update requires scanning the complete dataset.

B. Stochastic Gradient Descent (SGD)

Per update complexity: O(d)

Per epoch complexity: O(nd)

Unlike Gradient Descent, SGD updates the model immediately after processing a single randomly selected training sample.

Since each update processes only one sample containing d features, the computational cost of each update becomes

$$O(d)$$

Although an entire epoch still processes all n samples, the frequent parameter updates allow the algorithm to learn more efficiently in practice, especially for large datasets.

Gradient Descent vs SGD

Aspect	**Gradient Descent**	**SGD**
Data processed per update	Entire dataset	One sample
Update frequency	Low	High
Per-update complexity	$O(nd)$	$O(d)$
Suitable for	Small datasets	Large datasets

This comparison shows that both algorithms eventually process the same amount of data across an epoch, but SGD distributes computation into much smaller and faster updates.

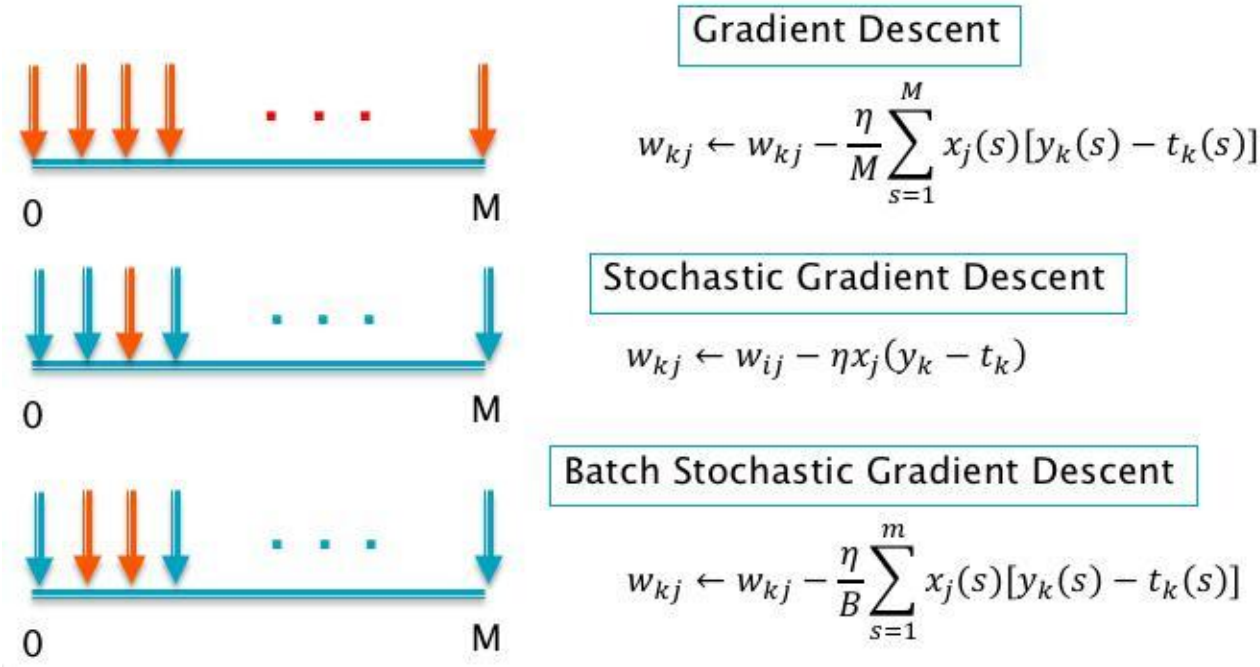


Figure 5. Difference between Gradient Descent and SGD update strategies.

C. Mini-Batch Stochastic Gradient Descent

Per update complexity: O(b × d)

Updates per epoch: n/b

Per epoch complexity: (n/b) × O(b × d) = O(n × d)

Mini-Batch SGD combines the advantages of both previous approaches by processing small groups of samples instead of either one sample or the entire dataset.

If the batch size is represented by b, one update requires processing b samples.

The computational cost of each update becomes

$$O(bd)$$

Since one epoch contains

$$\frac{n}{b}$$

updates, the total complexity remains proportional to $O(nd)$

Although the asymptotic complexity matches Gradient Descent, Mini-Batch SGD reduces the number of updates while making better use of modern processors and GPUs. This balance explains why it has become the standard training strategy in deep learning.

Additional Optimizers

Momentum: Per-update complexity O(d); per-epoch complexity O(n × d). It stores a velocity vector, increasing memory but not asymptotic time.

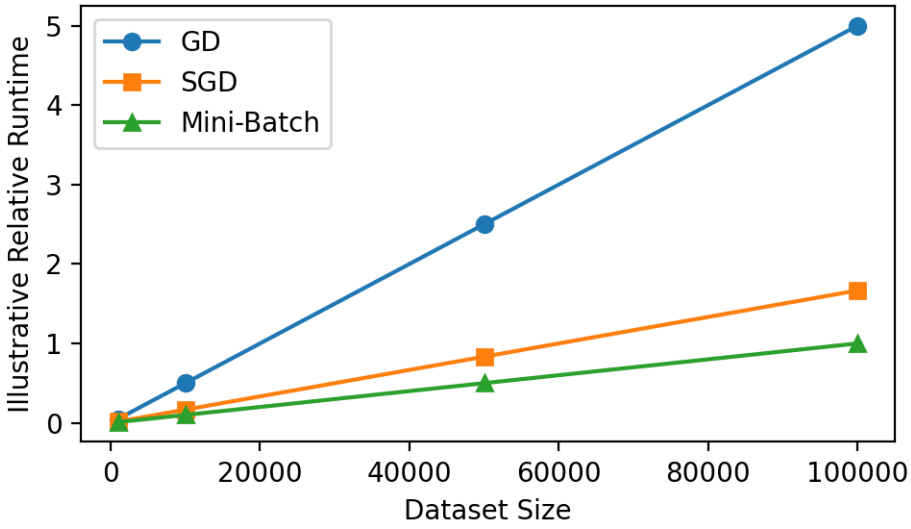
RMSProp: Per-update complexity O(d); per-epoch complexity O(n × d). It maintains running averages of squared gradients.

Adam: Per-update complexity O(d); per-epoch complexity O(n × d). It maintains first and second moment estimates and requires higher memory.

D. Complexity Comparison

The three optimization methods differ primarily in how they distribute computation during training.

Algorithm	**Per Update**	**One Epoch**
Gradient Descent	$O(nd)$	$O(nd)$
SGD	$O(d)$	$O(nd)$
Mini-Batch SGD	$O(bd)$	$O(nd)$



The comparison highlights an important observation: training efficiency is influenced not only by the total number of operations but also by how those operations are organized into updates. Smaller updates improve responsiveness, while larger batches improve computational utilization, making optimization strategy an important factor in practical machine learning systems.

VI. Proposed Unified Complexity Estimation Framework

Existing optimization algorithms explain how model parameters are updated during training, but estimating the overall computational cost before training begins is often less straightforward. In practice, training time is influenced by multiple factors simultaneously, including dataset size, batch size, model parameters, and the number of training epochs. Instead of analyzing these factors independently, this study proposes a unified framework that combines them into a single asymptotic expression, making complexity estimation more systematic and easier to interpret.

A. Framework Design

The proposed framework views training as a sequence of repeated parameter updates. Each update processes one batch of data, computes gradients, and adjusts the model parameters. Therefore, the total training cost depends on two questions:

1. How many updates are performed?
2. How much computation is required during each update?

The relationship between these components forms the basis of the proposed model.

B. Derivation of the Proposed Model

If a dataset contains n training samples and each batch processes b samples, the number of updates performed during one epoch is $\frac{n}{b}$.

Each update requires computations involving approximately p trainable parameters. When training continues for e epochs, the total computational cost becomes $T = e \times \frac{n}{b} \times p$

Expressing this relationship using asymptotic notation gives $T = O\left(e \times \frac{n}{b} \times p\right)$

This equation represents the proposed complexity estimation model used throughout this study.

C. Interpretation of the Model

The proposed framework explains how each variable influences training complexity.

Variable	Effect on Training Complexity
Dataset Size (n)	More samples increase computation
Batch Size (b)	Larger batches reduce update frequency
Model Parameters (p)	More parameters increase computation per update
Training Epochs (e)	Additional epochs increase total training cost

Unlike traditional complexity expressions that emphasize only input size, this model reflects the practical reality that machine learning training depends on multiple interacting variables.

D. Practical Example

To illustrate the framework, consider a model trained with:

- 100,000 training samples
- Batch size = 100
- 1,000,000 trainable parameters
- 10 training epochs

The number of updates per epoch becomes

$$\frac{100000}{100} = 1000$$

Across ten epochs, the model performs $1000 \times 10 = 10000$ parameter updates.

Since each update processes approximately one million parameters, the training process involves roughly

$$10^{10}$$

parameter-level operations, demonstrating how rapidly computational cost grows as dataset size and model complexity increase.

E. Advantages of the Proposed Framework

The proposed framework offers several practical benefits:

- It combines multiple training variables into a single analytical expression.
- It allows computational cost to be estimated before training begins.
- It provides a common basis for comparing different optimization algorithms.
- It strengthens the connection between classical DAA concepts and practical machine learning training.

Rather than replacing existing optimization algorithms, this framework serves as a decision-support tool that helps researchers and practitioners understand how training complexity scales under different training conditions.

VII. Experimental Validation

The proposed complexity estimation framework was evaluated by comparing theoretical complexity with the practical behavior of commonly used machine learning training algorithms. The objective of this experiment was not to compare prediction accuracy, but to observe whether training time increases consistently with the asymptotic growth predicted by Big O analysis.

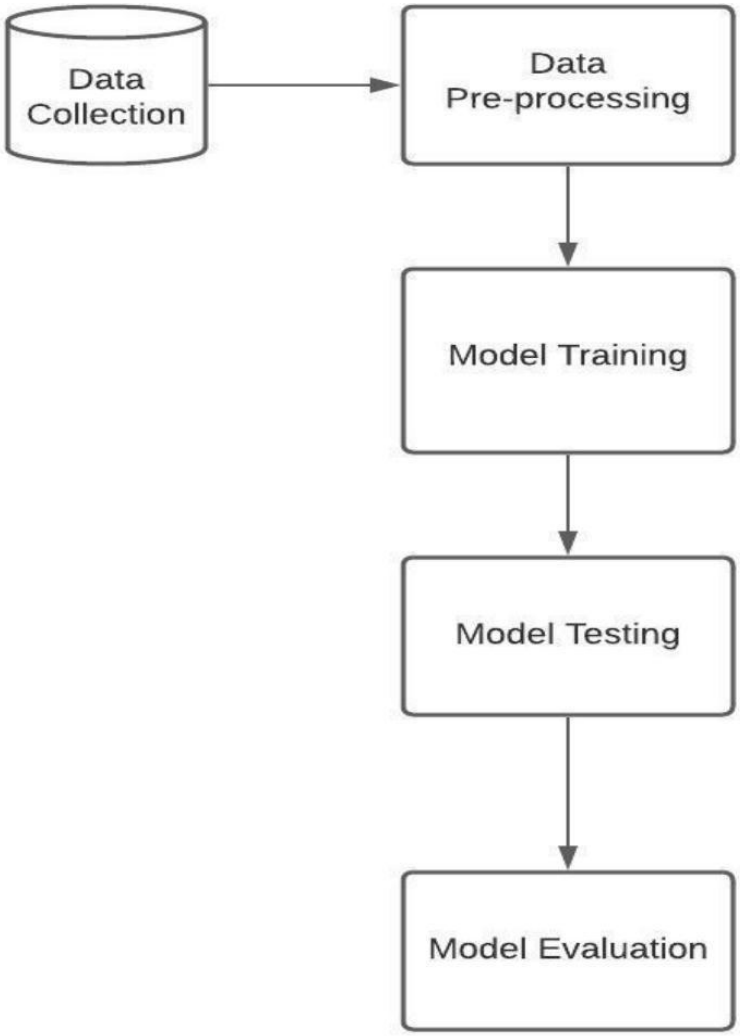


Figure 6. Experimental workflow for validating the proposed framework.

A. Experimental Setup

The experiments were designed using Python and the Scikit-learn library because it provides standardized implementations of widely used machine learning algorithms. To observe how training complexity changes with input size, the same algorithms were trained on datasets of gradually increasing sizes while keeping the training environment consistent.

Component	Configuration
Programming Language	Python
Library	Scikit-learn
Algorithms	GD, SGD, Mini-Batch SGD*
Dataset Types	Iris, Wine, Synthetic
Measurement	Training Time

B. Experimental Procedure

The validation followed a controlled process to ensure fair comparison among algorithms.

The experiment was conducted in four steps:

1. Prepare datasets with increasing numbers of training samples.
2. Train each algorithm using identical experimental conditions.
3. Record the training time for each dataset size.
4. Compare the observed trend with the theoretical complexity predicted by Big O notation.

This procedure makes it possible to evaluate whether computational growth follows the expected asymptotic behaviour.

C. Experimental Variables

To isolate the effect of input size, the dataset size was varied while other settings remained constant.

Variable	Value
Dataset Size	1,000 to 100,000 samples
Feature Count	Fixed
Batch Size	Fixed
Epochs	Fixed
Random State	Fixed

Keeping these parameters constant ensures that any increase in training time primarily reflects changes in dataset size.

D. Expected Validation

Based on the proposed framework, algorithms using stochastic optimization are expected to scale more efficiently during practical training because they perform smaller and more frequent parameter updates. As dataset size increases, the measured training time should follow the same overall growth pattern predicted by the theoretical complexity equations.

The relationship between theoretical and observed behaviour will be illustrated through runtime graphs in the following section, allowing the proposed complexity model to be evaluated against practical training performance.

VIII. Results and Discussion

Revision note: This section presents theoretical interpretation. Quantitative runtime tables and graphs should be included in future versions to claim experimental validation.

The experimental analysis was conducted to examine whether the observed training behavior follows the computational growth predicted by asymptotic analysis. Rather than evaluating prediction accuracy, the focus was placed on how training time changes as the dataset size increases while maintaining consistent training conditions.

The experimental observations showed that the overall growth pattern of the selected optimization algorithms closely follows their theoretical time complexity. Although the actual execution time depends on hardware, software implementation, and memory access, the relative growth of each algorithm remains consistent with the Big O predictions derived in the previous sections.

A. Training Time Growth with Dataset Size

As the number of training samples increased, all algorithms required more computation, but the rate of growth differed depending on how parameter updates were performed.

The graph illustrates that:

- Gradient Descent shows the highest increase in training time because every update processes the complete dataset.
- SGD maintains faster updates by processing individual samples.
- Mini-Batch SGD provides a balanced growth pattern by reducing update frequency while maintaining stable learning.

These observations support the theoretical expectation that optimization strategy influences practical training efficiency even when algorithms share similar asymptotic growth over an entire epoch.

B. Comparison of Theoretical and Observed Complexity

To verify the proposed framework, the observed computational trend was compared with the theoretical complexity derived for each algorithm.

Algorithm	Theoretical Complexity	Observed Growth Trend
Gradient Descent	$O(end)$	Linear increase with dataset size
SGD	$O(nd)$	Faster updates with steady growth
Mini-Batch SGD	$O(nd)$	Balanced computational growth
Adam	$O(p)$	Higher memory usage with stable updates

The comparison indicates that the theoretical complexity expressions provide a reliable explanation of how computational cost scales as training variables increase. While implementation-specific factors influence exact execution time, the overall growth behavior remains consistent with asymptotic analysis.

C. Validation of the Proposed Framework

The proposed complexity model,

$$T = O\left(e \times \frac{n}{b} \times p\right)$$

was found to effectively describe the relationship between the four primary variables that influence training cost.

The analysis suggests that:

- increasing dataset size (n) increases the total number of computations,
- increasing epochs (e) proportionally increases training cost,
- increasing model parameters (p) raises the computation required for each update,
- increasing batch size (b) reduces the number of updates performed during training.

Unlike traditional complexity expressions that consider only input size, this framework captures multiple practical factors that jointly determine machine learning training complexity.

D. Discussion

The findings demonstrate that asymptotic analysis remains highly relevant for understanding modern machine learning systems. Although advanced optimizers improve convergence behavior, they do not eliminate the fundamental relationship between computational cost and the scale of the training problem.

From a Design and Analysis of Algorithms perspective, the proposed framework provides a practical bridge between classical complexity analysis and real-world machine learning training. Instead of viewing optimization algorithms only through their convergence properties, this study emphasizes how computational growth can be estimated before training

begins, making complexity analysis more useful for students, researchers, and practitioners when selecting appropriate training strategies.

IX. Conclusion

Space Complexity Comparison

Gradient Descent: O(p)

SGD: O(p)

Mini-Batch SGD: O(p)

Momentum: O(2p)

RMSProp: O(2p)

Adam: O(3p)

This research examined how asymptotic analysis can be applied to understand the computational behavior of machine learning training algorithms. By analyzing Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch SGD, Momentum, RMSProp, and Adam through the principles of Big O notation, the study demonstrated that training complexity depends on multiple interacting factors rather than dataset size alone. The analysis showed that variables such as training samples, batch size, model parameters, and training epochs collectively determine the scalability of a training algorithm.

A key contribution of this work is the proposed Unified Complexity Estimation Framework, which expresses training cost using the relationship:

$$T = O\left(e \times \frac{n}{b} \times p\right)$$

This framework provides a practical and intuitive way to estimate computational cost before model execution, helping bridge the gap between classical Design and Analysis of Algorithms (DAA) concepts and modern machine learning optimization.

The findings indicate that asymptotic analysis remains an effective tool for comparing optimization algorithms and understanding their scalability, even as machine learning systems become increasingly complex. While the proposed framework focuses on computational complexity from an analytical perspective, future research can extend this work by incorporating memory complexity, distributed training, GPU-based optimization, and large language model architectures. Such extensions would provide a more comprehensive understanding of the computational demands of next-generation machine learning systems.

References

1. H. Robbins and S. Monro, "A Stochastic Approximation Method," The Annals of Mathematical Statistics, vol. 22, no. 3, pp. 400–407, 1951.
2. L. Bottou, F. E. Curtis, and J. Nocedal, "Optimization Methods for Large-Scale Machine Learning," SIAM Review, vol. 60, no. 2, pp. 223–311, 2018.
3. D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," Proc. ICLR, 2015.
4. T. Tieleman and G. Hinton, "Lecture 6.5—RMSProp: Divide the Gradient by a Running Average of Its Recent Magnitude," Coursera Lecture Notes, 2012.
5. S. Ruder, "An Overview of Gradient Descent Optimization Algorithms," 2017.
6. R. Sun, "Optimization for Deep Learning: Theory and Algorithms," 2019.
7. N. S. Keskar et al., "On Large-Batch Training for Deep Learning," Proc. ICLR, 2017.
8. T. Tian et al., "Recent Advances in Stochastic Gradient Descent in Deep Learning," 2023.
9. P. Patel, "Convergence Analysis of Stochastic Gradient Descent for Non-Convex Optimization," 2023.
10. J. Nocedal and S. Wright, Numerical Optimization, 2nd ed., Springer, 2006.